

Equipment Rental Management System

Feasibility Study

Effat University • CS3151 Software Engineering • Fall 2026

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22 September 2026

1 General Information

The Equipment Rental Management System is a proposed Arabic/English web application for managing equipment availability, rental requests, rentals, returns, overdue items, maintenance, and payment records. The first intended user organization is my uncle's equipment rental business.

I selected this idea after interviewing my uncle. He explained that keeping track of equipment that is available, rented, overdue, or under maintenance can be difficult. A shared system could make records easier to maintain and reduce mistakes [3].

Faisal Alburti is the student preparing this study. The business owner is the interviewed uncle; his name and the business's registered name have not been provided. This is a student software project, not a newly established company. The existing business's legal form and ownership percentages are unconfirmed. No project equity partners or ownership allocations are proposed.

2 Executive Summary

We recommend developing a limited prototype and testing it with the owner and selected employees before operational use. The system will connect equipment and customer records with rentals, returns, payment entries, and overdue follow-up. Customers will be able to browse equipment and submit rental requests.

The core scope is the six agreed use cases: View Available Equipment, Rent Equipment, Return Equipment, Track Overdue Rentals, Update Equipment Status, and Process Payments. For the prototype, payment processing means recording payments received outside the system and updating balances. Live card processing is a future integration.

The proposed schedule is 12 weeks with 240 total student team hours. The illustrative first-year cash budget is SAR 1,650. A proposed SAR 2,000 funding allocation would cover this budget and retain a SAR 350 cash buffer. Funding is not committed, and student labor is unpaid in the cash forecast. Its economic value is separately estimated at SAR 7,200.

Technical and operational feasibility are conditional on team capability, accurate initial records, user acceptance, and successful testing. The financial benefit depends on time actually saved. One interview supports a practical need but does not prove wider market demand or commercial profitability.

3 Description of Products and Services

The system will serve three user groups identified in Lab 3: the business owner or manager, rental employees, and customers [2]. Its proposed value is a clear daily workflow with connected records and bilingual access. These features are not claimed to be unique in the market.

Use case	Main feature	Added value and user benefit
View Available Equipment	Search items, prices, and availability by rental dates.	Employees answer quickly; customers can choose suitable equipment.
Rent Equipment	Save customer details, period, agreed charges, and item.	A checked booking reduces conflicts and missing rental information.
Return Equipment	Record actual return date and equipment condition.	Staff close rentals accurately and flag damage or maintenance needs.
Track Overdue Rentals	List open rentals whose due dates have passed.	The owner and employees can follow up with the correct customer.
Update Equipment Status	Manage availability and maintenance records.	Unavailable equipment is kept out of rental selection.
Process Payments	Record amount, date, method, and remaining balance.	Staff and the owner can identify paid and unpaid rentals.

Supporting functions include customer rental requests, role-based access, rental history, basic management reports, and Arabic/English screens. A request remains pending until staff confirm availability. Damage or loss notes are recorded during returns; insurance claims and automated penalty decisions are outside the pilot.

4 Technological Impact

Positive effects include shared records, faster searches, consistent availability checks, and visible overdue balances. Negative effects include dependence on connectivity, the risk of incorrect data entry, security threats, and ongoing software maintenance. Training and backup arrangements are needed to manage these effects.

We propose a responsive web interface, server-side business rules, and a relational database. HTML, CSS, JavaScript, and a Python web framework are possible tools, subject to the team's skills. The team will build the prototype; hosting may be supplied externally. No provider or paid technology has been selected.

Future browser changes, security updates, and integration requirements may affect the application. We will keep the code modular, document the database, and avoid dependence on unnecessary integrations. GPS tracking, live payment gateways, automated SMS, and multiple branches will be deferred.

5 Country Economy

Saudi Arabia is the assumed geographic market. The historical indicators below provide context rather than evidence of demand for this particular system. The 2023 figure is from an IMF release; 2024 and 2025 figures are from the December 2025 Ministry of Finance budget statement. Different publication dates and statistical revisions limit direct comparison [4,5].

Year	Real GDP growth	Status of figure
2023	-0.8%	Historical estimate in the September 2024 IMF release [5].
2024	2.7%	Actual figure reported in the FY2026 budget statement [4].
2025	4.4%	Preliminary forecast in that statement, not a final 2025 result [4].

The FY2026 statement reports nominal GDP of SAR 4,703 billion for 2024 and estimates SAR 4,600 billion for 2025. It forecasts 4.6% real growth for 2026. These are dated planning figures, not a claim about the latest 2026 economic outcome [4].

Fiscal year	Revenue	Spending	Deficit	Data status
2024	1,259	1,375	116	Actual
2025	1,091	1,336	245	Updated estimate
2026	1,147	1,313	165	Approved budget

Amounts above are SAR billions, as reported and rounded by the Ministry of Finance; rounding can affect subtraction. The same statement reports stable-outlook ratings of A+ from S&P and Fitch and Aa3 from Moody's [4].

The published pattern is contraction in 2023 followed by recovery and forecast expansion. For purchasing power, the FY2026 statement estimates 2025 inflation at 2.3%; this does not establish a rental business's software budget [4]. Our inference is that affordability and operational value should be tested directly with the owner. Economic growth alone does not justify a sales forecast.

6 Product or Service Market

The immediate target is one equipment rental business. Potential later users are small and medium rental businesses that need simple records and bilingual access. Customers of those businesses need clear availability, prices, and return information. The interview identifies a problem at one business; it is not a representative market survey [3].

Lab 1 compared Booqable, EZRentOut, Rentman, and Current RMS. It found that existing platforms already provide many rental functions [1]. Spreadsheets, paper records, and separate payment records are alternatives. Future competitors could include local software vendors and wider business-management platforms.

Market size and purchasing behavior

The total Saudi rental-software market, the share supplied locally, and the share supplied by foreign vendors have not been established. Physical equipment import totals would not measure this software market. The serviceable pilot market is one business, not an estimate of national demand. No local-versus-imported percentage is assumed.

Historical software demand, rental transaction volumes, and the balance between supply and demand require further evidence. We should collect the owner's rental volumes, overdue frequency, current record-keeping effort, and budget, then interview additional businesses before estimating a wider market.

Customers would choose equipment based on availability, suitability, price, and rental terms. A business would choose our system if it makes daily tasks easier at an acceptable total cost. Arabic/English support and a focused workflow are proposed advantages; competitor language limitations and a lower price have not been proven.

7 Marketing Strategy

The first step is a demonstration to the owner and employees using familiar equipment examples. A short pilot will let them test common rental tasks and suggest improvements. This direct approach matches a small initial user group better than a broad advertising campaign.

If the pilot succeeds, later promotion could use referrals, local business demonstrations, and a simple product page. The message would focus on clear availability, connected rental records, and bilingual use. No paid advertising or software subscription sales are included in the initial forecast. Wider marketing should follow evidence of willingness to pay.

8 Business Activity and Human Resources

The owner acts as the business contact. The student coordinator organizes requirements and reviews, while development and testing responsibilities are shared within the project team. The actual team size and named assignments are not confirmed. The following capacity plan assumes three students, one owner, and one employee; these are planning roles, not statements about the business's current staffing.

Period	Proposed people involved	Additional paid hires
First 3 months	3 students developing; owner and 1 employee reviewing.	0
First 6 months	Owner and 1 employee using the pilot; 1 designated support contact.	0 assumed
First year	Same operating roles; review support capacity before expansion.	0 assumed
Role	Responsibilities and required skills	
Coordinator and analyst	Requirements, scope, schedule, and documentation; communication and basic analysis skills.	
Developers	Screens, database, and workflows; programming, SQL, and version-control skills.	
Testing lead	Test cases, defect checks, and user testing; careful documentation and validation skills.	
Owner and employee	Confirm rental rules and opening records; practical rental experience and basic computer skills.	

9 Timeline

The proposed schedule starts after academic scope approval and agreement on access to business feedback. Twelve weeks and 240 total team hours are estimates. The team must align them with the actual course deadline. No funding, hiring, or launch commitment is assumed.

Timing	Milestone	Completion condition
Weeks 1–2	Requirements and operational plan	Owner reviews rental rules, scope, and acceptance criteria.
By week 2	Funding decision and role assignment	Agree budget source, team roles, and feedback availability; no paid hiring planned.
Weeks 3–4	Database and interface design	Review bilingual screens and linked records.
Weeks 5–7	Core implementation	Six main use cases work with test data.
Weeks 8–9	Requests, reports, and integration	Customer requests and access permissions work together.
Weeks 10–11	Testing and corrections	Check booking conflicts, balances, maintenance, permissions, and recovery.
Week 12	Trial operation and demonstration	Owner and employee complete representative tasks; document pilot results.
After pilot review	Operational launch decision	Proceed only if critical defects are fixed and support is agreed.

10 Technical Section

The application would be used at the existing rental premises and through customers' browsers. No new office, workshop, or machinery is required for the software pilot. The exact premises and network quality need confirmation; location is suitable if staff can access the system where rentals and returns occur.

Resource	Acquisition method or supplier
Computer and smartphone	Use existing business or student devices, subject to confirmation.
Development tools and database	Use suitable open-source tools; select versions after a team skills review.
Hosting, domain, and backup storage	Choose a provider after checking cost, access, recovery, and hosting requirements.
Equipment and customer data	Obtain necessary records from the owner with permission; verify opening balances and active rentals.

Each equipment item receives a unique ID. Staff verify availability before confirming a rental. Returns record condition; damaged equipment is blocked until cleared. Overdue status is calculated from an open rental's due date. Payment records remain linked to the rental, and edits should be traceable.

Use authentication, password hashing, role permissions, input validation, and encrypted hosting access. Customers see only their own records. Before real use, confirm applicable privacy, licensing, VAT, and invoicing obligations. ZATCA describes generation and integration phases for e-invoicing [6]; a prototype payment receipt must not be assumed to satisfy them. Municipal fees for the existing shop are outside this software budget; changes affecting the premises should be reviewed by the owner.

11 Business Model

The initial model is an internal management system funded by the existing rental business if the owner approves. The business continues earning revenue by renting equipment. The software's expected contribution is better record control and staff productivity; additional rental revenue is not assumed.

Element	Proposed approach
Delivery	A web application used by staff, with a basic customer browsing and request page.
Distribution	Provide staff accounts and a customer web link; physical pickup and return remain at the business.
Pricing	The owner sets equipment rates and rental terms. No software subscription price is set for the pilot.
Value beyond profit	Clearer records, fewer conflicts, better follow-up, and an easier rental experience.
Possible later commercialization	Consider a subscription or setup-and-support model only after market and support-cost validation.

12 General Risk Factors

Risk or characteristic	Possible effect	Response
Limited team time and experience	Late delivery or workflow defects.	Keep the scope small; review progress weekly and test core flows early.
Inaccurate opening records	Incorrect availability or balances.	Reconcile items and active rentals with the owner before pilot use.
Simultaneous booking attempts	The same item is promised twice.	Check conflicts when saving and test concurrent attempts.
Staff adoption and missed entries	Records become unreliable.	Provide brief training, clear responsibility, and a daily review.
Data loss or unauthorized access	Business disruption or exposure.	Test permissions and recovery; minimize customer data collected.
Competition and economic change	Low willingness to pay; higher costs.	Measure user value and compare the total cost of alternatives.
Changes in applicable rules	Extra development or operating work.	Review official requirements before launch and budget integrations separately.

13 Capability

The team must demonstrate basic web development, SQL, testing, and documentation skills through an early rental-and-return prototype. Course participation alone does not prove these skills. The owner supplies rental knowledge and validates business rules; the students supply implementation work.

Existing devices and open-source tools make the proposed resource plan plausible, but device access, hosting suitability, and available team hours need confirmation. Business experience comes from the stakeholder rather than from an assumed professional development history. Funding and long-term support are not yet committed. If capability is insufficient, reduce scope or evaluate an existing platform before proceeding.

14 Financial Expectations

All project figures are illustrative estimates in Saudi riyals, not quotations or the uncle's actual accounts. The model covers the software pilot only. It assumes existing devices and internet, unpaid student development, and no software sales. Actual rental-business revenue, equipment assets, payroll, and profit are outside this forecast.

Budget and financing assumptions

Cost item	Timing or basis	SAR
Setup and training	One time	500
Contingency	One time; assumed fully spent in this model	250
Hosting and domain	Annual allowance	600
Backup provision	Annual allowance	120
Minor maintenance expenses	Annual cash allowance; excludes developer wages	180
First-year cash expenditure	750 initial + 900 recurring	1,650
Unpaid development effort	240 team hours × SAR 30 per hour	7,200
First-year economic cost	Cash expenditure + development time value	8,850

A proposed owner allocation of SAR 2,000 would fund the first year and leave SAR 350 unspent. A further SAR 900 per year is assumed for years 2 and 3. This is a funding scenario, not a promise from the owner. Recurring developer labor and major upgrades require a separate support estimate.

First year quarterly income and expenditure

Q1 includes deployment and the initial SAR 750 cost. Recurring costs are modeled evenly at SAR 225 per quarter; actual invoices may fall differently. Time benefits start after deployment. No interest, tax, or Zakat charge is allocated to this internal project model; this is not an assessment of the existing business's obligations.

Measure in SAR	Q1	Q2	Q3	Q4	Year 1
Software sales revenue	0	0	0	0	0
Initial expenses	750	0	0	0	750
Recurring expenses	225	225	225	225	900
Project cash-basis result	-975	-225	-225	-225	-1,650
Owner funding allocation	2,000	0	0	0	2,000
Closing project cash	1,025	800	575	350	350

Owner funding is financing, not revenue. The negative project result reflects internal expenditure. It does not mean that the rental business itself is losing money. All amounts are assumed paid when incurred, with no receivables, unpaid bills, debt, or new capitalized assets.

Three year income and cash flow projections

Measure in SAR	Year 1	Year 2	Year 3
Software sales revenue	0	0	0
Cost of software sales	0	0	0
Gross result before project expenses	0	0	0
Initial and recurring project expenses	1,650	900	900
Net project result before unallocated taxes	-1,650	-900	-900
Operating cash flow	-1,650	-900	-900
Investing cash flow	0	0	0
Financing cash flow from owner	2,000	900	900
Net cash movement	350	0	0
Opening cash balance	0	350	350
Closing cash balance	350	350	350

Illustrative project financial position

Year-end position in SAR	Year 1	Year 2	Year 3
Current assets consisting of cash	350	350	350
Fixed assets	0	0	0
Total assets	350	350	350
Current and long-term liabilities	0	0	0
Cumulative owner contributions	2,000	2,900	3,800
Cumulative project expenditure	-1,650	-2,550	-3,450
Net project equity or funding balance	350	350	350
Total liabilities and equity	350	350	350

Assets equal liabilities plus the remaining funding balance in each year. These are simplified project statements, not statutory accounts or the balance sheet of a new company. Existing business equipment is excluded, and unpaid student time is not a cash payment or recognized asset in this model.

Cost and benefit analysis

Assume 20 staff hours are saved each month, valued at SAR 30 per hour: $20 \times 30 \times 12 = \text{SAR } 7,200$ per year. This is potential productive time, not automatic cash savings or new sales. After cash costs, modeled benefits are SAR 5,550 in year 1 and SAR 6,300 in each later year. After also deducting SAR 7,200 of development effort, the three-year economic benefit is SAR 10,950.

The initial economic investment is SAR 7,950. Monthly modeled benefit after SAR 75 operating costs is SAR 525, giving economic payback of about 15.1 months after deployment. At five hours saved per month, payback is about 106 months. With no measured savings, the economic case is unsupported. Continued use depends on owner funding and verified benefits, not on assumed software profit.

15 Capital Requirements and Strategy

When required	Proposed source and use	SAR
Before deployment	Owner funding: first-year costs of 1,650 plus cash buffer of 350.	2,000
Start of year 2	Owner funding for annual operating costs.	900
Start of year 3	Owner funding for annual operating costs.	900
Three-year total	3,450 forecast spending + 350 closing cash.	3,800

The proposed source is an internal allocation from the existing rental business, subject to the owner's decision. The model assumes no bank borrowing, outside investors, pledged equipment, or grant approval. No software revenue or asset sale is used to secure financing. The proposed debt-to-equity funding ratio is 0:1 because all financing is an owner contribution.

There are no investor distributions in the pilot. Expected returns are operational value and potential time savings. Under the assumptions in Section 14, the three-year economic return on total economic cost is $10,950 \div 10,650 \times 100 \approx 102.8\%$. This is an undiscounted scenario measure, not a cash investment return. The SAR 350 unspent buffer is not an expense and is excluded from that calculation.

16 Results and Recommendations

The project is conditionally feasible as a focused student prototype. Its strength is that it addresses a problem described by an equipment rental owner and supports the owner, employee, and customer workflows developed in the earlier labs. Its main limitations are untested adoption, unverified savings, limited development time, and no committed support or financing.

Dimension	Conclusion
Technical	Proceed if the team demonstrates the core workflow and tests data integrity, permissions, and recovery.
Operational	Proceed if the owner verifies initial records and staff can reliably perform daily tasks.
Schedule	The 12-week estimate requires the assumed team effort and strict scope control.
Economic	Potentially worthwhile under the modeled time savings; confirm costs and measure benefits during the pilot.
Market	One stakeholder need is supported; wider demand, pricing, and market share remain unvalidated.

We recommend building the six main use cases first, then adding the basic customer request page, bilingual screens, and reports. Before operational launch, representative users should complete the main tasks, overlapping bookings should be blocked, maintenance items should be unavailable, payment balances should reconcile, and customers should see only their own records.

The team should also demonstrate backup restoration, resolve critical defects, and agree ongoing responsibility for maintenance. If these conditions cannot be met within the approved scope and budget, reduce the pilot or compare an existing rental platform. The evidence supports testing the proposal; it does not justify a guaranteed success rate.

References

[1] Equipment Rental Management System, Market Study – Lab 1, CS3151, Fall 2026. Supplied project submission. Its competitor findings are secondary research, and it states that no stakeholder interviews were conducted for that study.

[2] CS3151 Lab 3 Persona. Supplied project submission containing owner, employee, and customer personas, feature mapping, scenarios, and user stories. Persona names are fictional design examples, not confirmed interview participants.

[3] Faisal Alburti’s account of an interview with his uncle, an equipment rental business owner. The interview identifies difficulties tracking availability, rentals, overdue items, and maintenance. No interview date or financial figures were supplied.

[4] Saudi Ministry of Finance. Budget Statement FY2026, December 2025. Pages 31, 34, 51, and 75. Figures retain the status and reporting date of this publication. [Open source](#)

[5] International Monetary Fund. IMF Executive Board Concludes 2024 Article IV Consultation with Saudi Arabia. 3 September 2024. Historical 2023 growth indicator. [Open source](#)

[6] Zakat, Tax and Customs Authority. E-Invoicing overview. Accessed 22 September 2026. [Open source](#)

Appendix A Assignment Template Coverage

The study uses the PDF model's 16 main sections in order. The table below shows where the Word template's nine sections are included. The PDF's repeated customer-benefit prompt is answered once in Section 3. Manufacturing and commercial-launch prompts are adapted to software delivery and pilot operation.

Word template section	Location in this study
Executive Summary	2
Description of Products and Services	3
Technology Considerations	4 and 10
Product or Service Marketplace	6
Marketing Strategy	7
Organization and Staffing	8
Schedule	9
Financial Projections	14 and 15
Findings and Recommendations	16